PRESENTATION

STATS PROJECT

DATA ANALYSIS

A TAKE ON BEHAVIOUR AND ECONOMICS

31 OCTOBER, 2025

The First Scenario - The Laser Peril

The first question in our game — where we asked you to pick 2 out of 4 cards to test the rule “If a card has a vowel on one side, it must have an even number on the other side” — is called the Wason Selection Task.

This task shows something interesting about how people think. Most of us naturally try to confirm what we believe instead of trying to disprove it. This is called confirmation bias.
In our card example, many people pick A and 4, because they try to check if the rule is true. But the rule never said that an even number must have a vowel behind it.

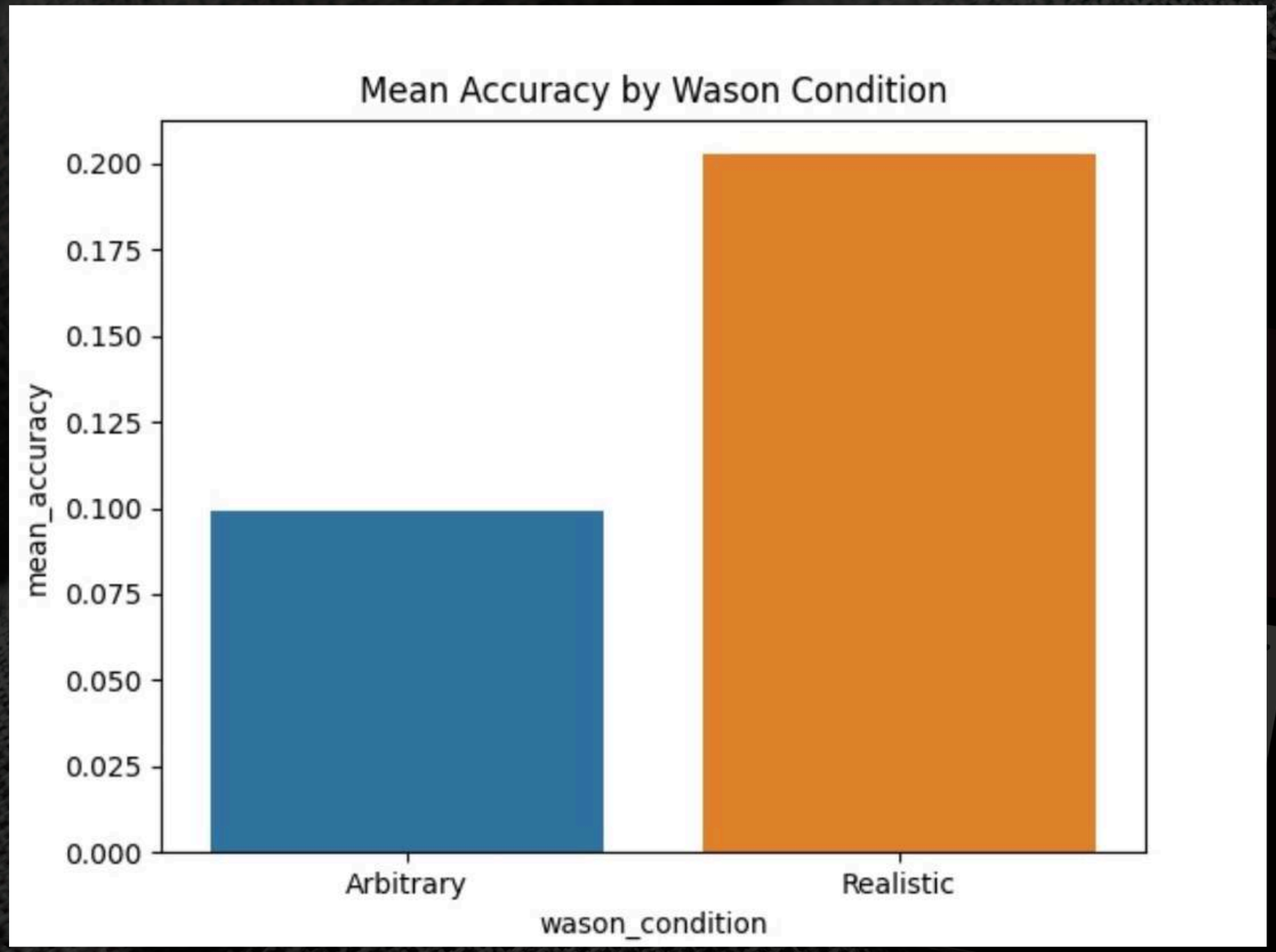
The correct cards to check are A and 7.

- You check A to see if it has an even number behind it (which would support the rule).
- You check 7 because if there's a vowel behind it, the rule would be proven wrong.

Our data looks at how accurate people are in these types of situations — both in abstract, made-up conditions like the card rule and in realistic conditions, such as “If a person is 18 years old, they can drink alcohol.”

We also study how factors like reaction time and other behaviours affect how accurate people are. the Wason task shows how our minds tend to confirm our beliefs instead of looking for what might prove us wrong.

WASON SELECTION TASK

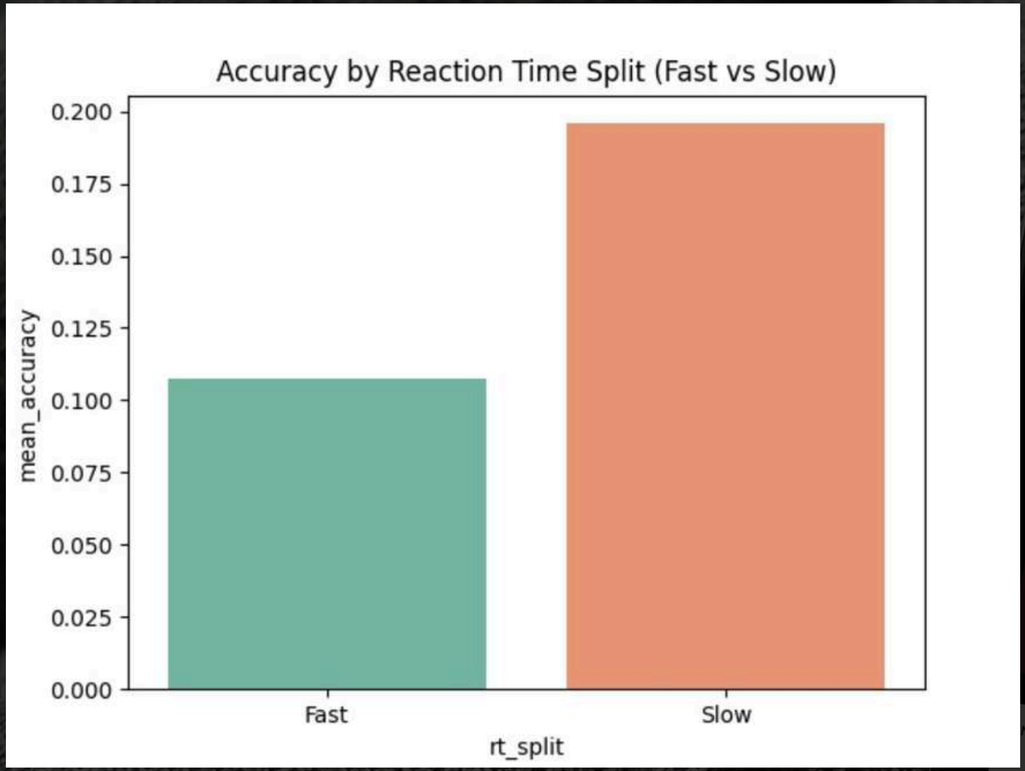


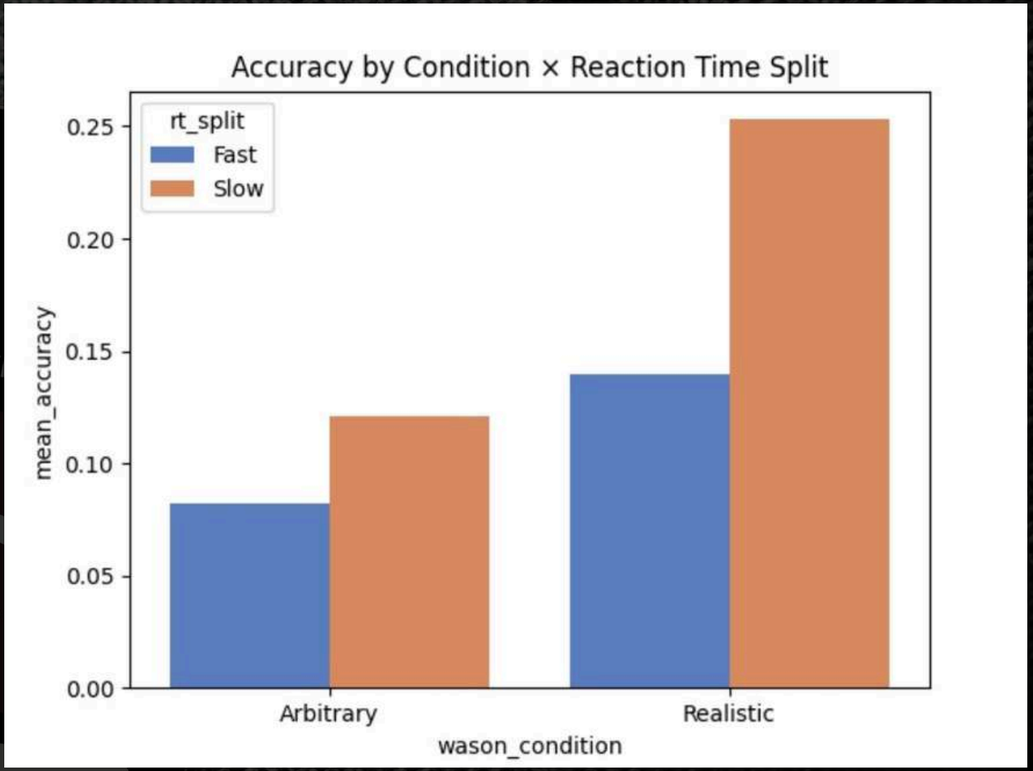
MEAN ACCURACY

Wason Condition	Mean Accuracy
Arbitrary	0.09894
Realistic	0.202749

Real-world context improves reasoning — accuracy in the Realistic condition (0.20) was about 2× higher than in the Arbitrary condition (0.10).

WASON





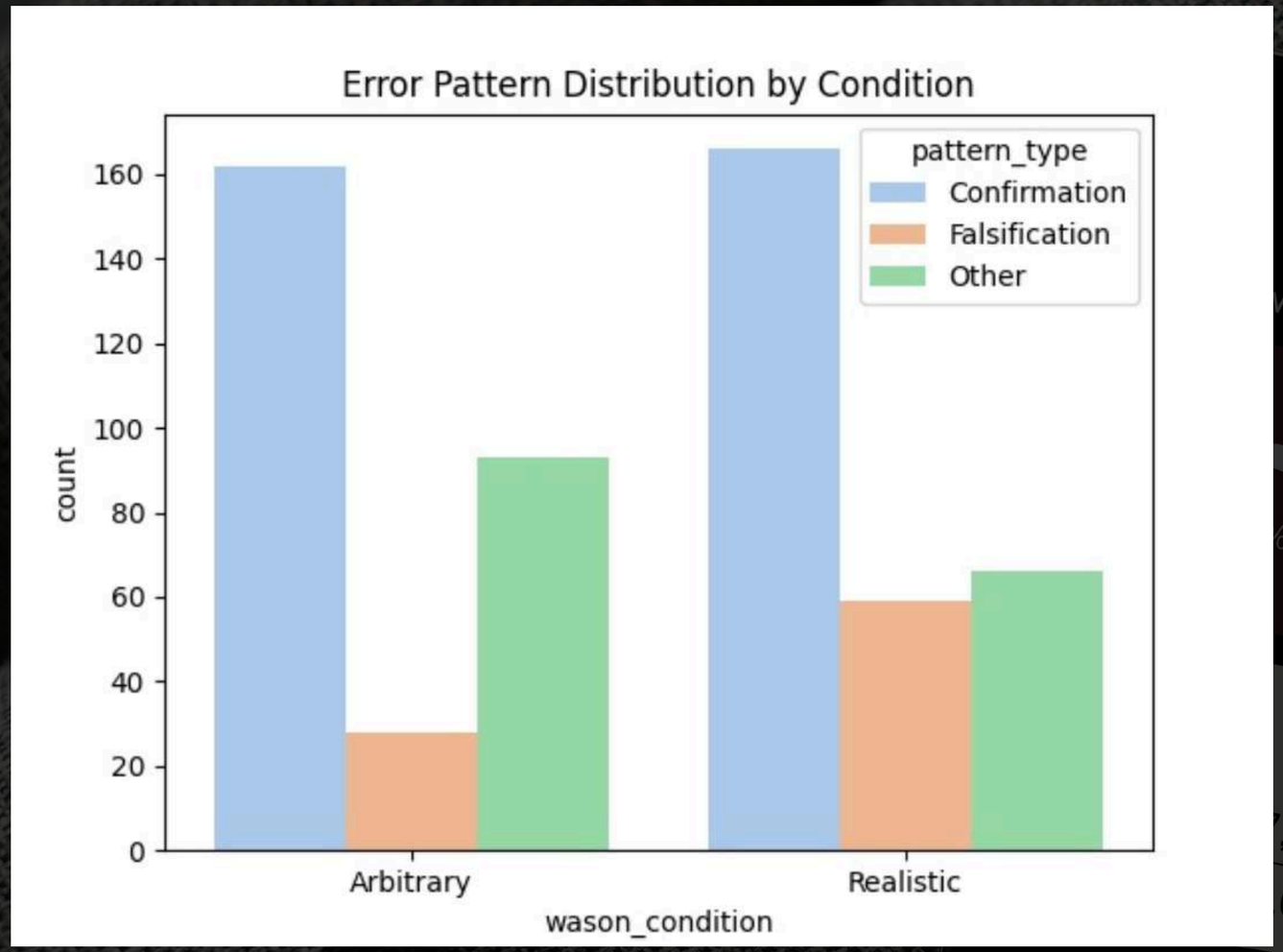
Response Time	Mean Accuracy
Fast	0.107639
Slow	0.195804

Slower responses led to higher accuracy, suggesting thoughtful reasoning helped

Wason Condition	Response Time	Mean Accuracy
Arbitrary	Fast	0.081761
Arbitrary	Slow	0.120968
Realistic	Fast	0.139535
Realistic	Slow	0.253086

Accuracy improves with both realistic context and slower responses, showing deeper reasoning under these conditions.

WASON



Most participants showed a confirmation bias, but realistic scenarios led to more falsification reasoning, indicating better logical thinking.

Wason Condition	Pattern Type	Count
Arbitrary	Confirmation	162
Arbitrary	Falsification	28
Arbitrary	Other	93
Realistic	Confirmation	166
Realistic	Falsification	59
Realistic	Other	66

WASON

KEY REGRESSION RESULTS

Variable	Coefficient	Effect	P | z |	Significant?
Intercept	-3.144278	0.04x	0.00000000..	True
Wason Condition	0.7894727	2.20x	0.001747243	True
Bonus	0.4944727	1.63x	0.06220182	Slightly Unreliable
rt(s)	0.1	1.1x	0.0000957	True

The regression shows that realistic condition, bonus (incentive) , and response time had significant positive effects on accuracy.

WASON

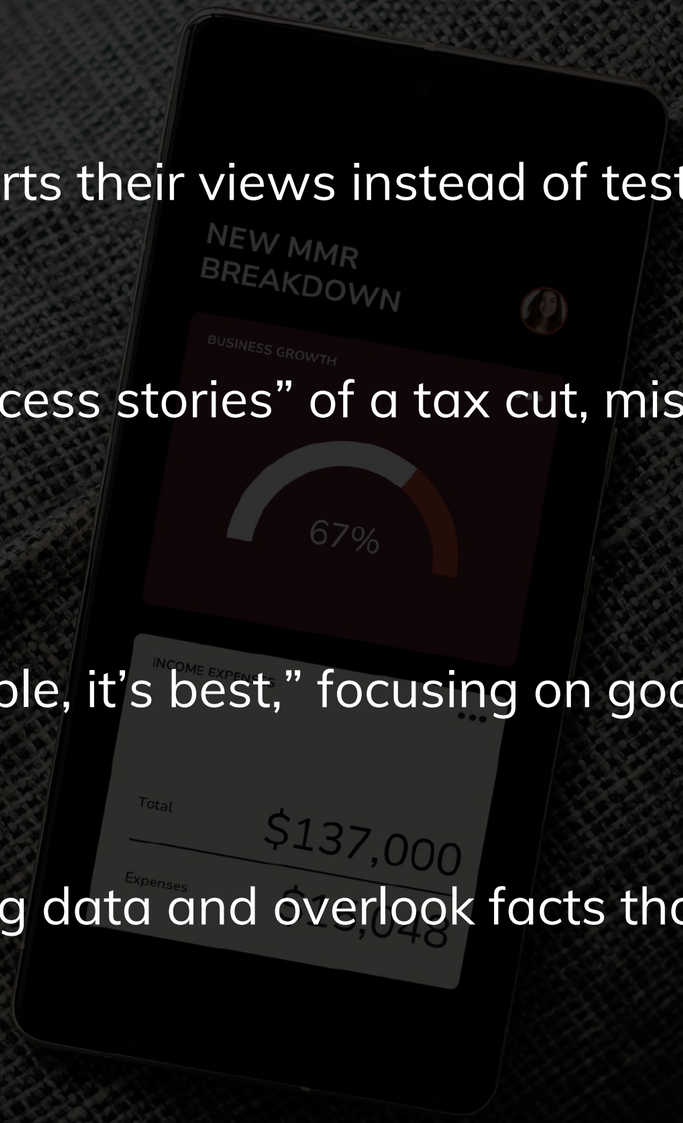
ECONOMICS INTERPRETATION

Investor Confirmation Bias
Investors often seek news that supports their views instead of testing if they're wrong.

Policy & Government Decisions
Policymakers may track only the "success stories" of a tax cut, missing where it failed — a classic Wason error.

Consumer & Brand Loyalty
Consumers follow rules like "If it's Apple, it's best," focusing on good reviews and ignoring faults.

Market Experiments
Studies show people prefer confirming data and overlook facts that could disprove their beliefs.



The Second Scenario - Nuclear Meltdown

The Next 3 numeric questions were a part of the test called the **Cognitive Reflection Test** or **CRT** for short, these tests are meant to reveal intuitive bias in people, where rather than stopping and thinking logically people make errors by choosing intuitively. This is often seen in the real world as well where people tend to underestimate exponential growth and fall into debt traps. Our analysis shows that most people often start out in thinking intuitively and make errors but once they start thinking logically they keep on doing so and make less mistakes.

1) Drone & Battery Price Puzzle (Simple & Humanized)

You spend 1.1 credits to buy a drone and its battery.
The drone costs 1 credit more than the battery.
What is the price of the drone and the battery?

Correct Answer	Intuitive Answer
Drone = 1.05	Drone = 1
Battery = 0.05	Battery = 0.1

2) Machines Making Chips (Clear & Realistic)

If 5 machines can make 5 chips in 5 days,
then how many days will 10 machines take to make 10 chips,
assuming all machines work at the same speed?

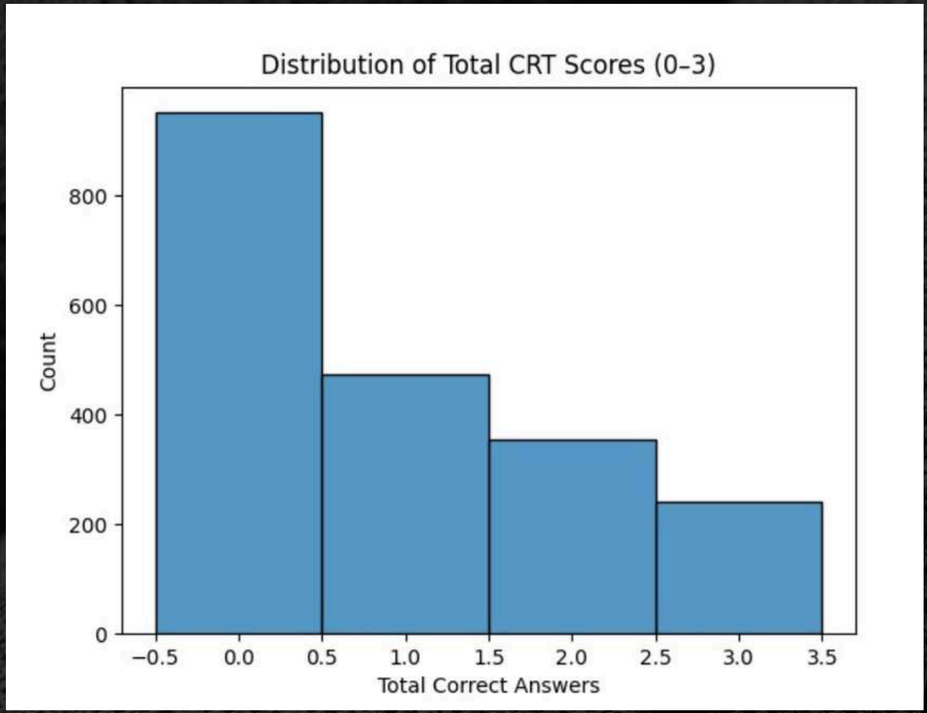
Correct Answer	Intuitive Answer
5 Days	10 Days

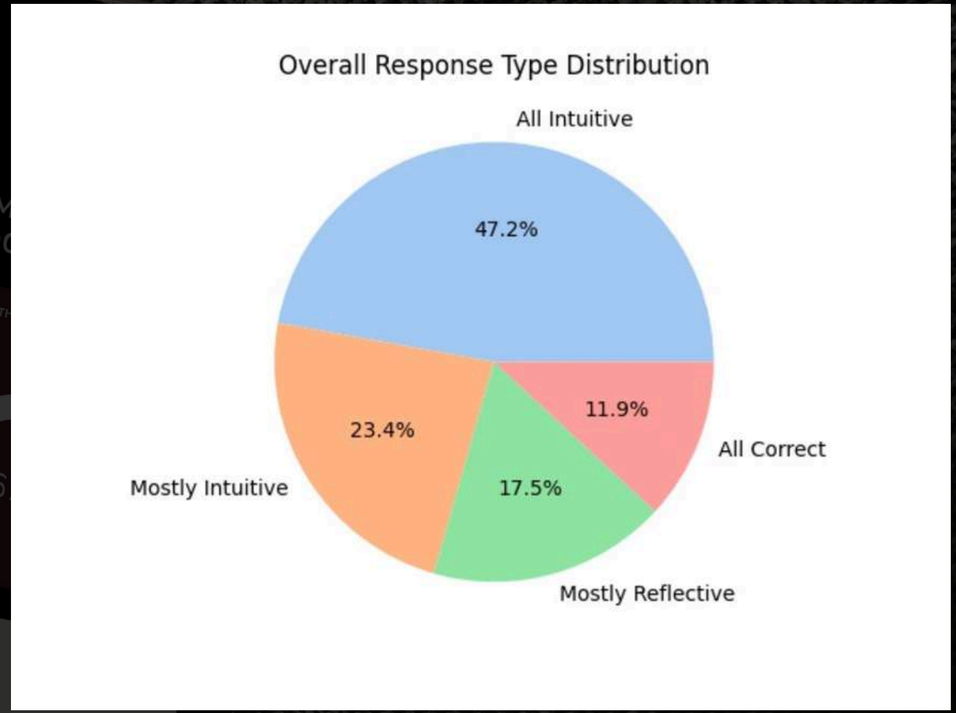
3) Virus Growth in a Computer (Simple & Humanized)

A computer virus doubles in size every day.
The computer becomes fully infected on day 48.
On which day was the computer 50% infected?

Correct Answer	Intuitive Answer
47 Days	24 Days

RESPONSE TYPE DISTRIBUTION(CRT):





DISTRIBUTION OF TOTAL CRT SCORES

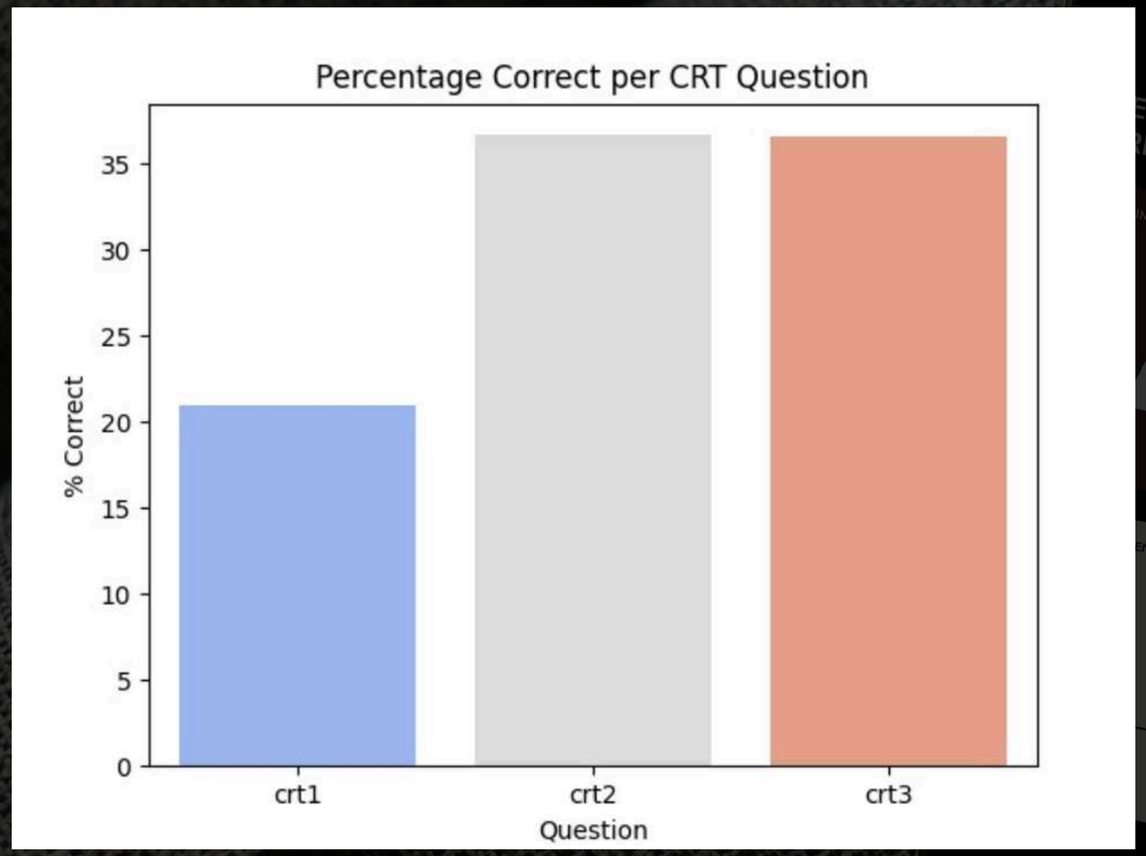
Category	Count
All Intuitive	952
Mostly Intuitive	473
Mostly Reflective	354

OVERALL RESPONSE TYPE DISTRIBUTION

Total Score	Percentage
0	47.15%
1	23.43%
2	17.53%
3	11.89%

RESPONSE TYPE DISTRIBUTION(CRT):

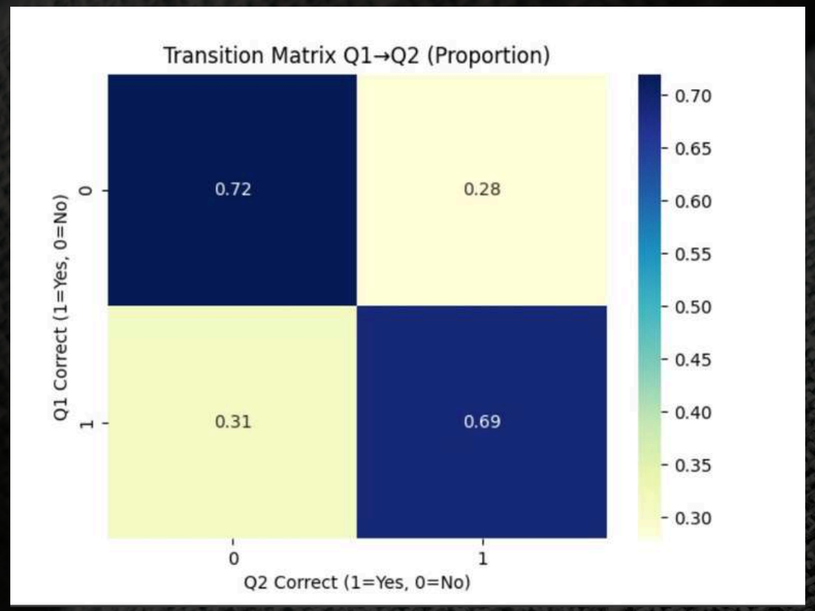
PERCENTAGE OF CORRECT RESPONSES PER QUESTION

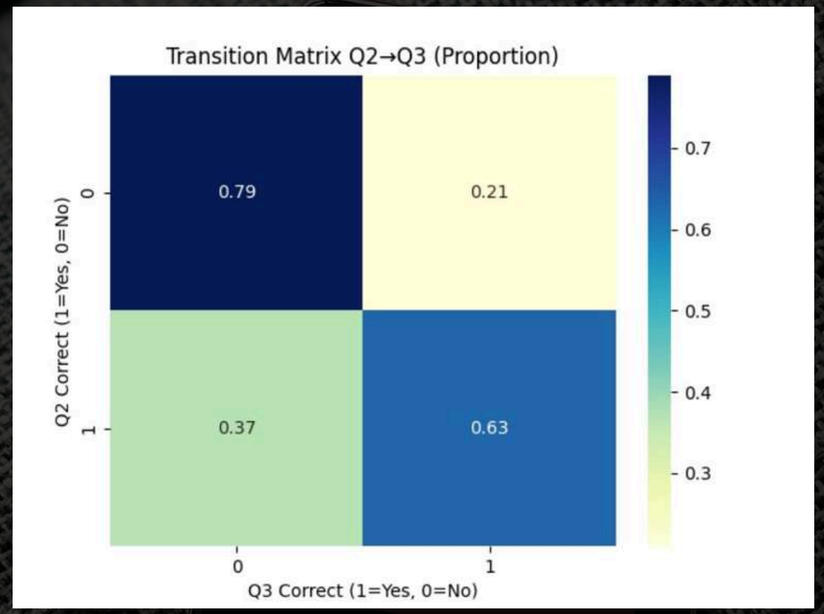


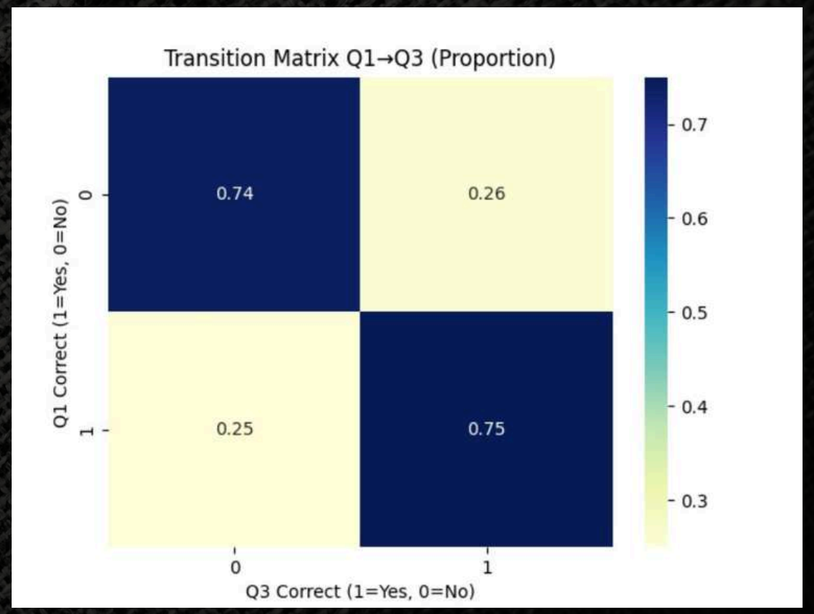
Question	Percentage Correct
crt1	20.90%
crt2	36.65%
crt3	36.60%

CRT2 and CRT3 show nearly identical performance levels, while CRT1 had a significantly lower correct response rate — suggesting it was more challenging or prone to intuitive errors.

RESPONSE TYPE DISTRIBUTION(CRT):







- 72% got Q1 wrong — 31% of them got Q2 right.
- 28% got Q1 correct — 69% of them got Q2 right

- 79% got Q2 wrong — 37% of them got Q3 right.
- 21% got Q2 correct — 63% of them got Q3 right.

- 74% got Q1 wrong — 25% of them got Q3 right.
- 26% got Q1 correct — 75% of them got Q3 right.

THE LOGISTIC RUN:

LOGISTIC REGRESSION SUMMARY: Q2_CORRECT ~ Q1_CORRECT

Metric	Value	Metric	Value
Dependent Variable	crt2	No. of Observations	2019
Model Type	Logit	Method	MLE
Df (Model)	1	Df (Residuals)	2017
Pseudo R-squared	0.08917	Log-Likelihood	-1208.3
LL-Null	-1326.6	LLR p-value	0.000000000..
Converged	TRUE	Date	Tue, 28 Oct 2025
Time	21:58:46	Optimization Iterations	5
Current Function Value	0.526579		

Variable	Coefficient	Std. Error	z-value	P>|z|	[0.025	0.975]
Intercept	-0.9419	0.056	-16.909	0	-1.051	-0.833
crt1	1.7511	0.119	14.685	0	1.517	1.985

Getting Q1 right increases odds of Q2 success by ~5.7× (exp(1.7511)).

THE LOGISTIC RUN:

LOGISTIC REGRESSION: Q3_CORRECT ~ Q1_CORRECT + Q2_CORRECT

Metric	Value	Metric	Value
Dependent Variable	crt2	No. of Observations	2019
Model Type	Logit	Method	MLE
Df (Model)	1	Df (Residuals)	2017
Pseudo R-squared	0.08917	Log-Likelihood	-1208.3
Converged	TRUE	LL-Null	-1326.6
Covariance Type	nonrobust	LLR p-value	0.000000000000

Variable	Coefficient	Std. Error	z-value	P>|z|	[0.025	0.975]
Intercept	-1.5406	0.074	-20.796	0	-1.686	-1.395
crt1	1.7031	0.134	12.739	0	1.441	1.965
crt2	1.5073	0.109	13.87	0	1.294	1.72

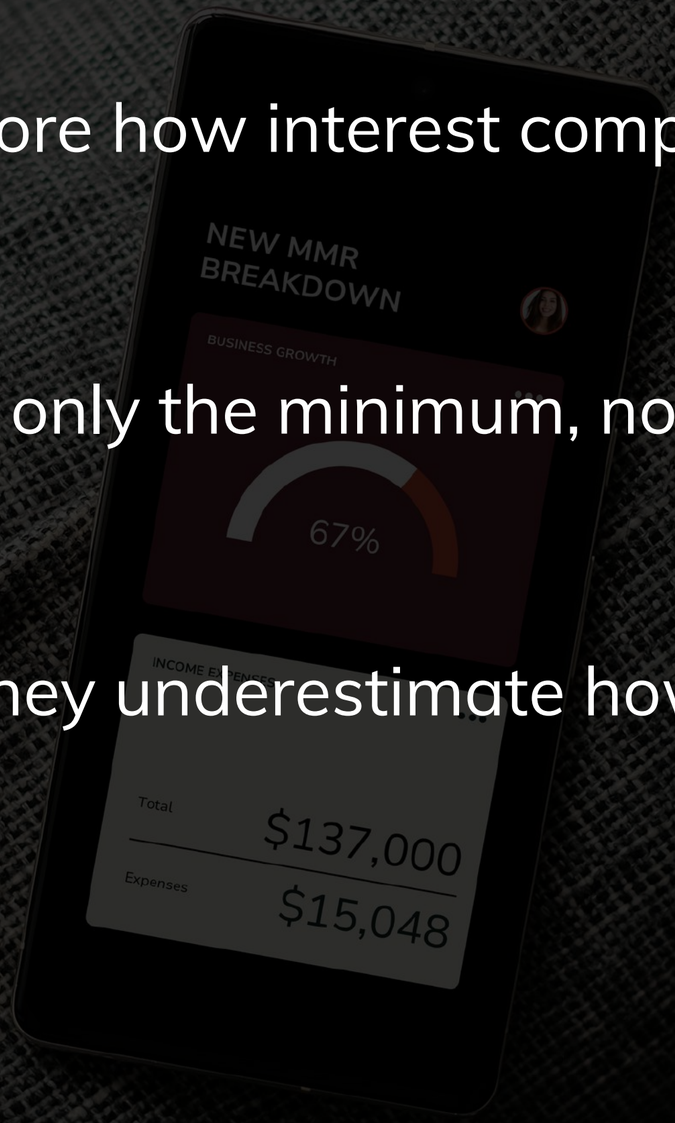
Getting Q1 right increases odds of Q3 success by ~ (exp(1.70) ≈ **5.5x**).

Getting Q2 right increases odds of Q3 success by ~ (exp(1.51) ≈ **4.5x**).

Getting Q1 and Q2 right increase the odds of Q3 success by ~**5.5 * 4.5 = 24.75x**

ECONOMIC INTERPRETATION

- Debt traps: Borrowers ignore how interest compounds, thinking payments grow slowly.
- Credit misuse: People pay only the minimum, not realizing debt doubles faster than expected.
- Poor investment timing: They underestimate how returns accumulate, missing long-term gains.



The Third Scenario - Oxygen Guess

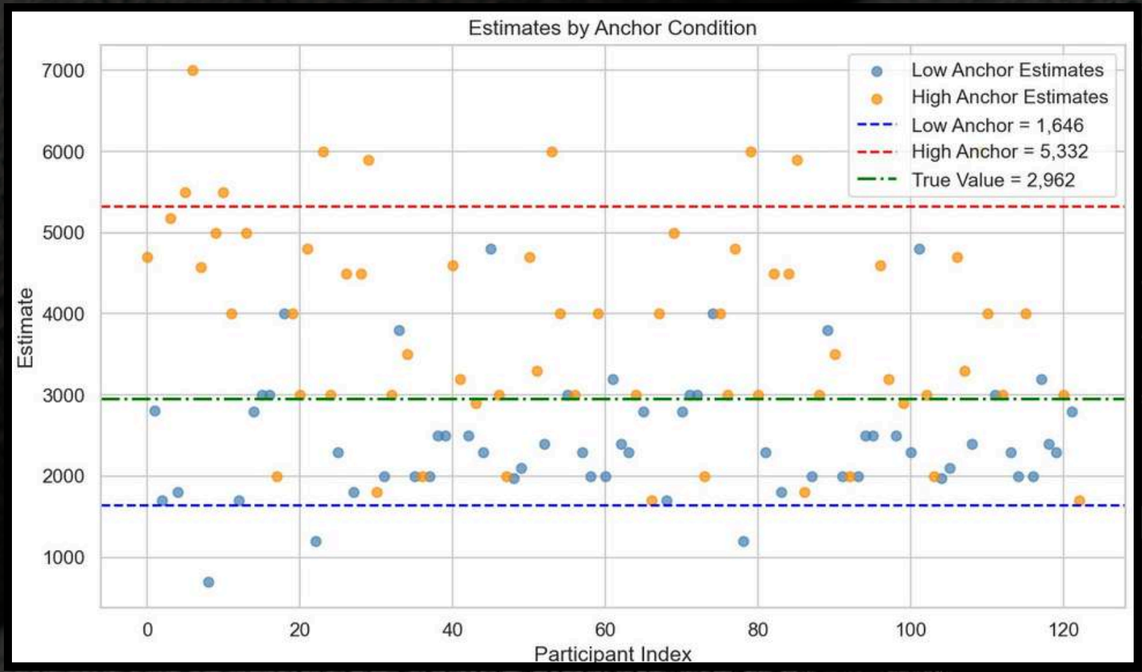
The next scenario illustrates a powerful decision-making bias known as the Anchoring Effect. Anchoring happens when people rely too heavily on an initial piece of information — the anchor — when forming judgments, estimates, or decisions. Even if the anchor is random or irrelevant, it can strongly influence the final answer people give.

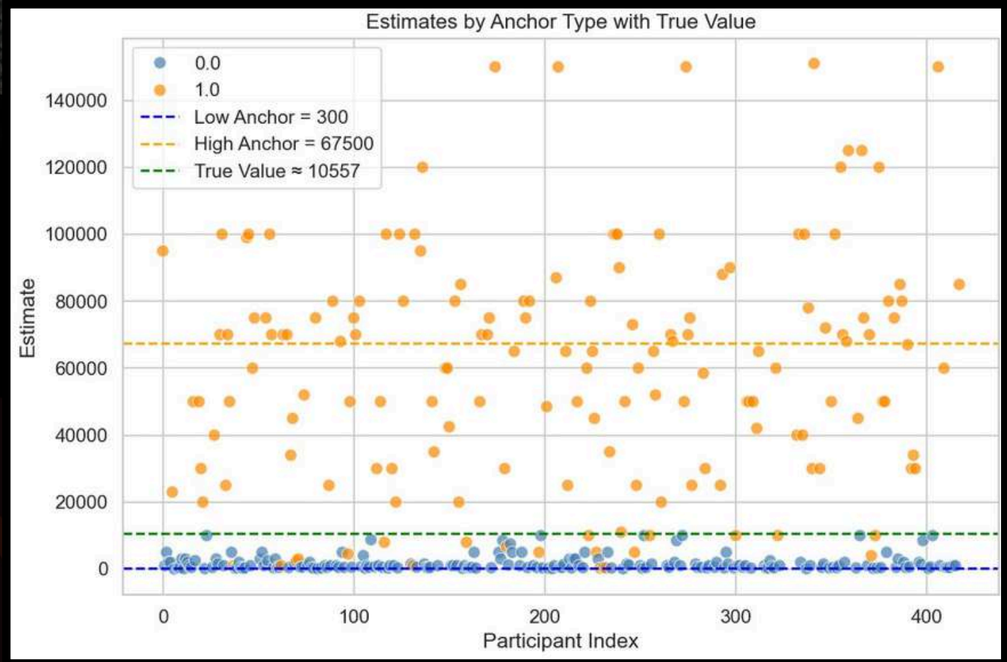
In our game, participants were asked to estimate the remaining amount of oxygen in a tank. Meanwhile, the number 70 was subtly flashing in the background. This was intentionally designed to bias their judgment, nudging their estimates toward the anchor without them consciously noticing it.

Our analysis shows that most people's guesses shifted toward the number anchors, demonstrating how anchors quietly shape decision-making. Once individuals become aware of this bias, however, they can adjust their thinking and make more accurate, deliberate choices.

Here we explore how people use anchoring in real life with several examples — in sales, price negotiations, salary discussions, and many other everyday situations. Anchoring plays a major role in shaping expectations, influencing perceived value, and steering the direction of negotiations. Understanding this effect helps individuals avoid falling for anchored judgments and make more rational decisions.

ANCHORING BIAS ACROSS DIFFERENT SCALES





MOUNTAIN HEIGHT ESTIMATE

Anchor–Estimate correlation: 0.540

Anchor Type	Mean Estimate	Mean Bias	Std Bias	Min Bias	Max Bias	N
Low Anchor	2472.75	-489.25	771.08	-2262	1838	60
High Anchor	3845.38	883.38	1306.67	-1262	4038	63

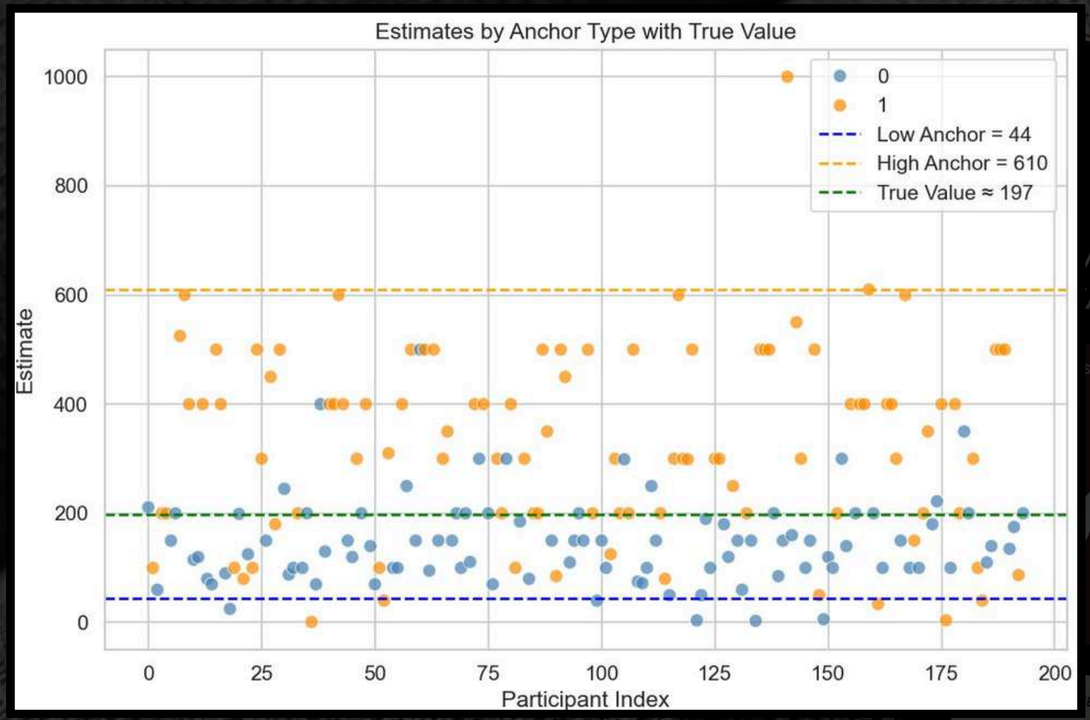
BIRTH RATE ESTIMATE

Anchor–Estimate correlation: 0.770

Anchor	Mean Est	Mean Bias	SD	N
Low	1705.68	-8851.32	2275.57	187
High	59108.52	48551.52	35157.43	155

In case of a general question (height estimate of a famous mountain), anchoring effect is smaller as compared to a question where people didn't have any concrete idea about the number (birth rate per day in usa)

ECONOMICS INTERPRETATION



WILLINGNESS TO PAY

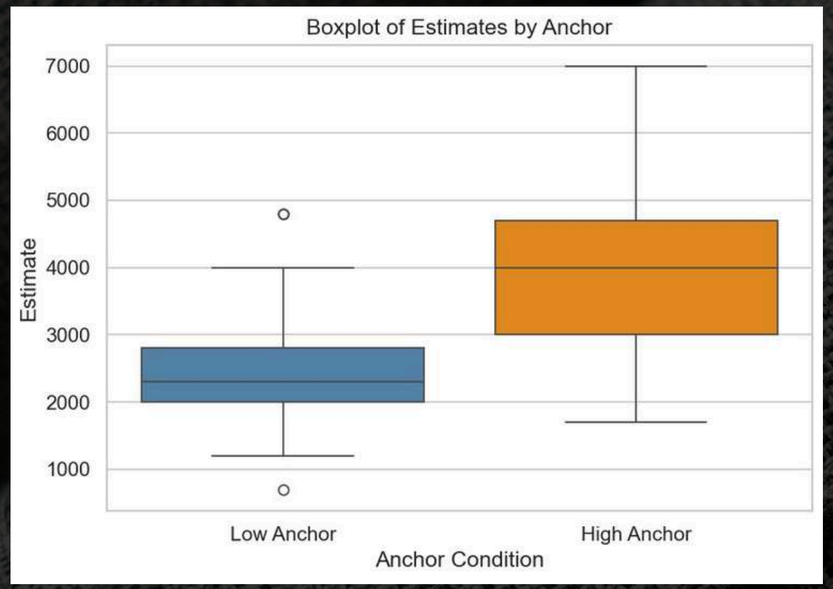
Anchor–Estimate correlation: 0.557

Anchor	Mean Est	Mean Bias	SD	N
Low	146.61	-50.67	81.1	96
High	330.1	132.82	176.07	98

- Apple iPad Launch: Jobs said $999 first, then $499 — high anchor made $499 look cheap, boosting sales.
- Retail Discounts: “Was ₹4999, now ₹2999” — old price anchors perception, making discount seem larger.
- Real Estate Pricing: Listing at ₹1.2 crore anchors buyers’ expectations, making ₹1 crore seem reasonable.
- Salary Negotiation: First salary offer sets an anchor that shapes final negotiation outcomes.

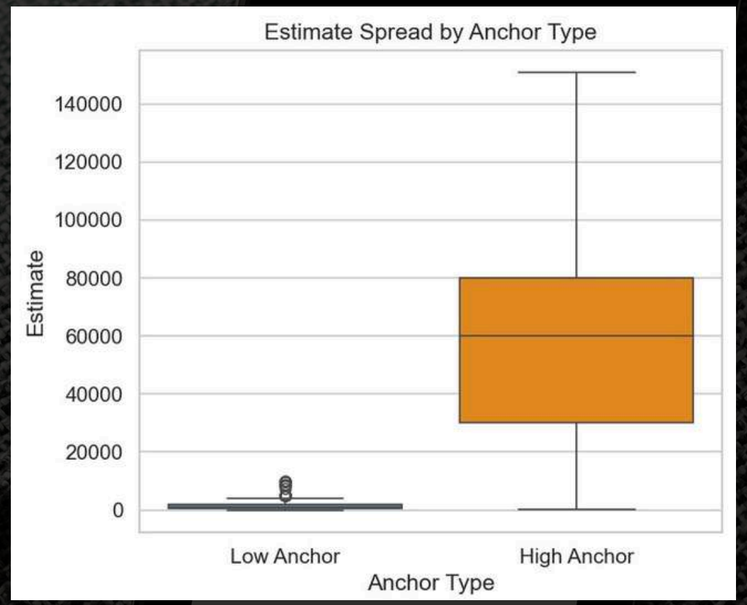
Steve Jobs Genius Marketing Startegy for IPads - Excellent example of Anchor effect ← Click to Watch

ANOTHER WAY OF VISUALISING - BOXPLOTS



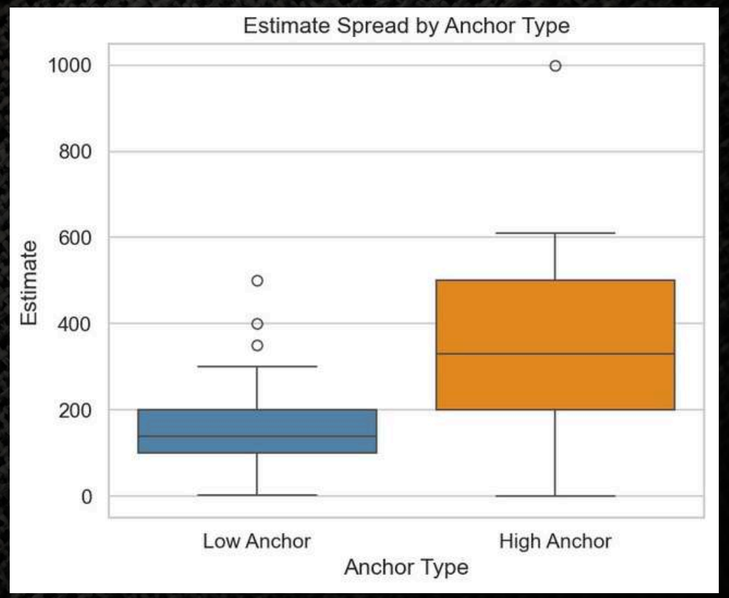
MOUNTAIN HEIGHT ESTIMATE

Anchor–Estimate correlation: 0.540



BIRTH RATE ESTIMATE

Anchor–Estimate correlation: 0.770



WILLINGNESS TO PAY

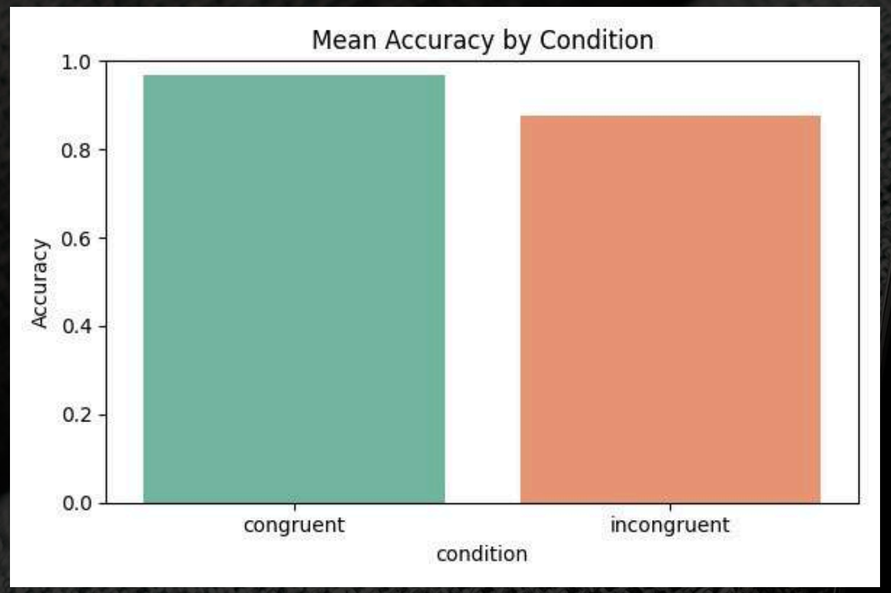
Anchor–Estimate correlation: 0.557

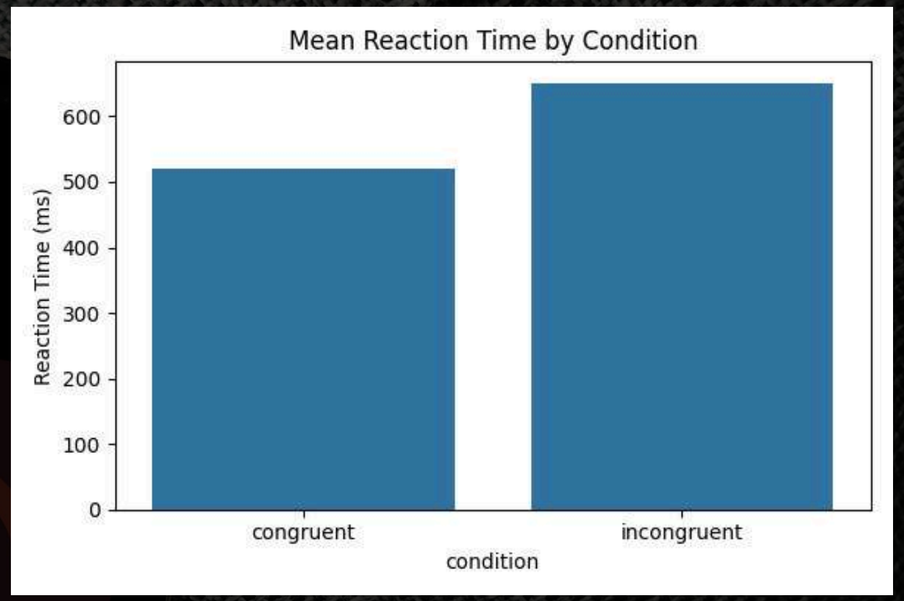
Scenario 4 - Colour Code Decode

This analysis relates to the colour–word interference question, commonly known as the Stroop Test.
A congruent condition occurs when the colour of the text and the meaning of the word are the same (for example, the word “RED” written in red). In these cases, people respond quickly and accurately because there is no conflict in processing.
However, in an incongruent condition — such as the word “RED” printed in blue — the brain receives two conflicting signals. This increases cognitive load, and the results show that both accuracy decreases and reaction time significantly increases.
The Stroop effect demonstrates how automatic processes, like reading, can interfere with our ability to perform simple perceptual tasks when conflicting information is present. Our analysis reveals that people in incongruent conditions take longer to answer and their accuracy falls as well.

Economically, the Stroop effect shows how confusing or look-alike cues can distort real-world decisions.
Just as conflicting colours slow us down in the task, similar packaging, slogans, or app icons create interference in markets — making people choose the wrong brand, trust misleading political messages, or even fall for fake digital services.
The same cognitive conflict that reduces accuracy in the Stroop test leads to irrational or mistaken choices in everyday economic environments

STROOP ANALYSIS





ACCURACY PER CONDITION

Condition	Accuracy
Congruent	0.9685
Incongruent	0.8748

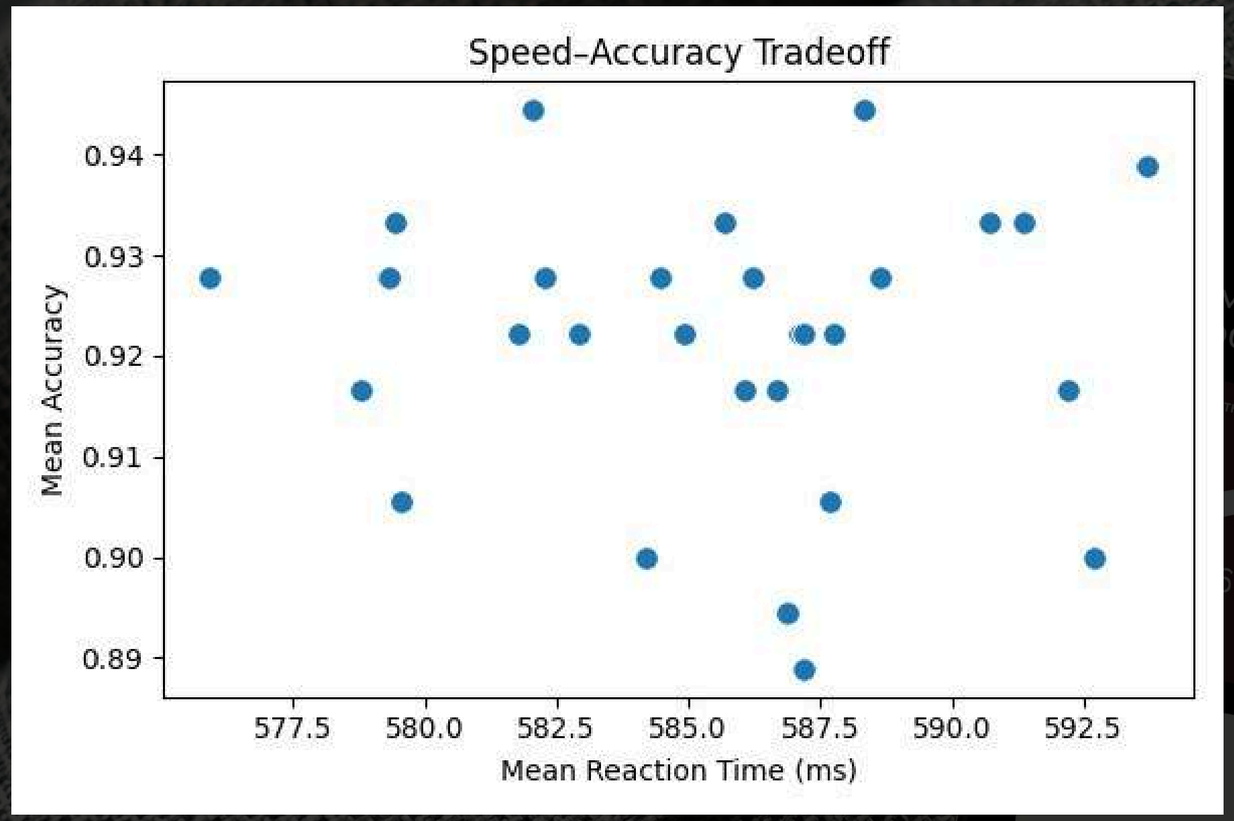
MEAN RT PER CONDITION

Condition	Reaction Time (ms)
Congruent	520.02
Incongruent	650.7

People make more accurate decisions when information is clear and consistent (like transparent prices or simple choices).
When information is conflicting, accuracy drops — much like misleading ads or unclear data can lead to incorrect financial or consumer decisions.

When information is straightforward, decisions are faster and more efficient — like quick market responses to clear news.
Confusing or mixed signals (as in the incongruent case) make people take longer to decide, reflecting slower market reactions and hesitation under uncertainty.

STROOP ANALYSIS



Speed–Accuracy Tradeoff (subject-level)

Subject	Mean RT	Mean Accuracy
1	585.69	0.9333
2	582.94	0.9222
3	587.76	0.9222
4	592.69	0.9
5	590.7	0.9333

Paired t-test Results

Measure	t-value	p-value
Reaction Time	-62.844	0
Accuracy	14.052	0

LOGISTIC REGRESSION

Variable	Coef	Std Err
Intercept	3.0683	0.454
C(condition)[T.incongruent]	-1.5725	0.167
reaction_time_ms	0.0007	0.001

STROOP ANALYSIS

ECONOMIC INTERPRETATION

- In advertisements: Brands use similar colors or packaging (like Bileri copying Bisleri) to trick consumers into trusting them.
- In policymaking: Political campaigns or welfare schemes use similar slogans or visuals to past successful ones → people associate them with old trusted programs.
- In digital markets: Fake apps or websites use similar icons or layouts as popular ones → users trust them and make poor financial choices.

CONCLUSION

HUMAN REASONING IS OFTEN BIASED, BUT REALISTIC CONTEXTS AND DELIBERATE THINKING IMPROVE ACCURACY — SHOWING THAT AWARENESS AND REFLECTION CAN LEAD TO BETTER DECISIONS IN BOTH DAILY LIFE AND ECONOMICS.

THANK YOU

FOR YOUR ATTENTION

SAKSHAM AGRAWAL
24B3021